Azure Data engineering (Udemy)

Resource Group:-

* It is a Container it stores related services

Azure DataFactory:-

1. Pipelines
2. Activities
3. Datasets
4. Linked services
5. Data Flow
6. Integration runtime.

Linked Services & Datasets:-

* Linked services are much like connection string, which define the connection information needed for the services to connecte to external resources.
* Dataset is a named view of data that simply points & refernces the data you want to use in your activities as inputs & outputs.
* Datasets identify data within different stores, such as tables, files, folders & documents.

Blob Storage:-

* Blob storage is optimized for storing massive amount of unstructured data.

Parameters :-

* Parameters are defined at the pipeline level, & can't be modified during a pipeline ~~level~~ run.
* Parameters are used to control behaviour of a pipeline & it's activities, such as by passing in the connection details for a dataset or the path of a file to be processed.

parameters → pipeline level
Dataset
Linked services.

JSON (Javascript object Notation) :-

* It stores data in the form of key value pair
* It is a lightweight format for storing & transporting data
* Data separated by commas.
* Curly braces hold objects
* Square brackets is arrays.

Integration runtime (IR) :-

* It is a compute infrastructure.

Types :-

1. Azure Integration runtime (Auto resolve IR) → It is managed by Azure.
2. self-hosted IR → It is managed by you → mainly for onpremise to cloud
3. Azure - SSIS IR → managed by Azure.

Copy activity:-

* Copy data from one location to another location.

Execute Pipline:-

* To execute another pipline. from one pipeline

Fail Activity:-

* we want to fail an pipeline or anything.

Get metadata Activity:-

* metadata is data about the data.

Lookup activity:-

* It reads & returns the content of a configuration file or table.
* It returns upto only 5000 rows
* It's size is must be less than 4MB otherwise It fails. the pipeline.

set variable activity:-

* It supports datatype string, boolean & array.

Wait activity:-

* Pipeline waits for the specified period of time.

Conditions

Switch activity:-

* Same as other programing language.

Filter activity:-

* To apply filter expression to an array type.
* Data must be array type from parameters or variable
* It takes Input only array it returns array.

For Each Activity:-

* Input format is array

Until activity:-

* ~~It is like~~ Condition is false it execute if condition is true it stops.

If Condition activity:-

* Same as othe programing language.

Conditional split:-

* It is under Dataflow
* By using it we can split Table into multiple tabs ~~using~~ by condition.
* It is present in Data flow.

Exists transformation:-

* Row filtering.
* By matching Column to store from left side details only it return left side Columns only.
* It is present in Data flow.

Union transformation:-

* To perform union, No. of Columns & corresponding Column data type must be same.
* It is present in dataflow

Join transformation:-

* Same like SQL
* It return both table columns. ~~This is the diff.~~
* It is present in Dataflow

Aggrate Transformation:-

* It is like a group by.
* It is present in dataflow

Surrogate Key transformation:-

* It creates a new Column with numeric representation with start & step to go to last Column assigns values.
* It is present in Data flow.

Select transformation :-

* Same like SQL
* By using it we can arrange ordering columns, drop & rename column
* It is present in data flow.

Lookup transformation :-

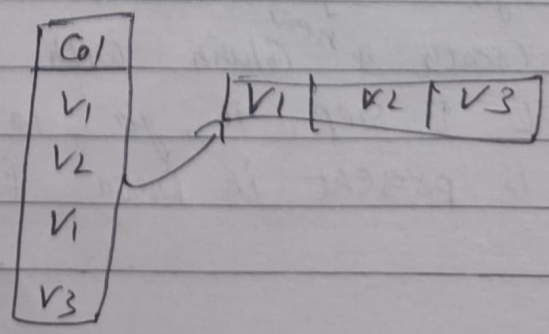
* It is similar to left outer Join
* It is present in data flow

Derived Column transformation :-

* By using it to generate new column in your data flow or to modify existing fields
* It is like a update statement & also we can create new column (withColum in pyspark)
* It is present in dataflow

Pivot Transformation :-

* By using it we can create multiple columns from the unique row values of a single column.
* It is present in dataflow.



Rank Transformation:-

* Same in SQL
* It is present in Dataflow

Window transformation:-

* To perform aggration on each row not like group by
In group by we get avg(sal) means we can avg but in window it changes every column value of window column.
* It is present in dataflow.

Flatten transformation:-

* Input file must be JSON file with array values.
* It is like an explode in databricks.
* It is present in data flow

Filter transformation:-

* It filters the rows based on the condition.
* It is just like filter function in python.
* It returns only true values not include false values.
* It is present in Data flow

Data warehouse :-

It is a Central repository of integrated data for analysis & reporting

Keys :-

Candidate Key :-

Unique + Not Null

Primary key :-

* one of the Candidate Key is a primary Key

Foreign Key :-

* An attribute in one table that refers to the primary key of another table.

Surrogate Key :-

* System generated unique identifier (usually auto-increment integer).

Slowly changing dimensions (SCDs) in Data warehousing

Dimension & Fact table :-

* Fact table stores measurable data
* Dimension table provide descriptive context

Why SCD?

a) Handling changes in Dimension Data

* Dimensions are not static, they change over time.
* Challange → How do we store history in data warehousing?

What is SCD?

* SCD are techniques to manage & track changes in dimension attributes over time.
* By using it preserves business history.

SCD types:-

Type 0 → Original data

Type 1 → Overwrite with New Data.

Type 2 → Keep full history (Able to store all history of data)

Type 3 → Keep partial history (Able to store past 2/3 history data)

Type 4 → History table (Separate table stores historical records)

Type 5 → Mini-Dimension

Type 6 → Hybrid (Combination of 1, 2 & 3)

Type 1:-

Ex: Merge Table 1 as T1
using Table2 as T2
on T1.Column = T2.Column
When matched Then
Update set T1.column = T2.Column, T1.Column = T2.Column
When not matched then
Insert Values (T2.Column, T2.Column, T2.Column);

With-Column :- Pyspark withColumn() is a transformation function of DataFrame which is used to

* Change data type of existing column in Dataframe.
* Adding new column to DF
* Changing the value of an existing column.
* Add a column with literal value
* Rename column name
* Drop DF column.

Ex:

df.withColumn("col-name", col("col_name").cast("Integer")). To change data type of column.

df = df.withcolumn("New_sal", col("salary")*1.1) To create new column.

df = df.withColumn("country", lit("India")) we have to use "lit" to fill defaut values to it.

df = df.withcolumn("sal", col("sal")+100) To update existing column values.

df = df.withColumnRenamed("mgr", "manager_Id"). By using withColumnRename we can change column name.

df = df.drop("columnname"). By using drop we can drop columns.

Delta Lake :-

* Data Lake + Data warehouse is called as Delta Lake.
* Delta lake is the optimized storage layer that provides the foundation for storing data & tables in the Databricks lakehouse

Data warehouse	Data Lake	Data Lakehouse / Delta Lake
* well structured Data works	* works well semi-structure & unstructured data.	* can handle structure, semi-structure & unstructured data
* mainly for BI & Analytics.	* Suitable for AI & ML	* suitable for both Data analytics & AML
* storage is costly & time consuming.	* storage is cost-efficient, fast & flexible.	* storage is cost-efficient, fast & flexable.
* ACID it follows.	* NON-ACID, so delete & update complex operation.	* It follows ACID properties
		* It is a combination advantages of Both Data warehouse & Data lake